## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

| | |
|---|---|
| **TITLE (PROVISIONAL)** | Community-Based Culturally Tailored Education Programs for Black Adults with Cardiovascular Disease, Diabetes, Hypertension, and Stroke: A Systematic Review Protocol of Primary Empirical Studies |
| **AUTHORS** | Fulton, Joseph; Singh, Hardeep; Pakkal, Oya; Uleryk, Elizabeth M; Nelson, Michelle |

## VERSION 1 – REVIEW

| | |
|---|---|
| **REVIEWER** | Raevschi, Elena<br>Nicolae Testemiceanu State Medical and Pharmaceutical University, Social Medicine and Health Management |
| **REVIEW RETURNED** | 14-Mar-2022 |

| | |
|---|---|
| **GENERAL COMMENTS** | The systematic review protocol title “Community-Based Culturally Tailored Education Programs for Black Adults with Cardiovascular Disease, Diabetes, Hypertension, and Stroke: A Systematic Review Protocol” is well formulated and it is informative. The aim of the proposed systematic review is clear and relevant. The background has been provided by adequate literature review including recent relevant studies and systematic reviews, as well. The authors clearly state the gap in the research which they are looking to fill rising clearly defined research questions. The study design is appropriate to answer two review questions stated in the protocol, and applied methods are described sufficiently to assure repeatability.<br>This systematic review will improve understanding on the design, structure and definition of community-based culturally tailored education programs (CBCT) that have been used between 2000-2021 to improve health outcomes in Black adults with cardiovascular disease, hypertension, diabetes, and stoke that is important for assuring more efficiency of future CBCT programs using Kreuter and colleagues framework. |

| | |
|---|---|
| **REVIEWER** | Goff, Louise<br>Kings College Lonodn, Diabetes and Nutritional Sciences |
| **REVIEW RETURNED** | 01-Apr-2022 |

| | |
|---|---|
| **GENERAL COMMENTS** | 'Community-Based Culturally Tailored Education Programs for Black Adults with Cardiovascular Disease, Diabetes, Hypertension, and Stroke: A Systematic Review Protocol'<br><br>I would ask the authors to consider the following comments/revisions:<br>[1] The title is incomplete and should include mention of the types of evidence being evaluated and the outcomes of interest. |

| | |
|---|---|
| | [2] Abstract - the authors make some strong claims in the abstract about the role of systemic racism and disparities in care. Whilst these are undeniably important drivers of health inequalities, there are a broad range of additional factors that contribute so i would encourage the authors to temper these statements a little, acknowledging that these are in part responsible.<br>[3] Title and methods - the review is focused on 'community-based' programs but how are they defining 'community-based' and how does this working definition compare/differ to definitions used by others in this field. This is a phrase that is used broadly and means different things to different people. I have the same comment regarding 'culturally tailored' - are the authors considering that 'culturally tailored' is distinct from 'community-based' i.e. could they have programs that are culturally tailored but not community-based and vice versa? Clear working definitions needed here.<br>[4] There are several frameworks available to evaluate culturally tailored strategies, can the authors briefly justify why they have chosen Kreuter's over the other options.<br>[5] Methods - the authors state 'the studies will be limited to articles published in English language'. Can they be clear if the SEARCH was limited or whether non-English studies were included in the search but excluded in the screening? Limiting the search has a very big effect, it is best to not limit the search.<br>[6] Top of page 10 is the following statement: 'we will include published empirical studies of any study design that gather evidence from a CBCT program..' It is not clear what sort of literature will be excluded - do the studies need to be an evaluation of a program on clinical outcomes or will literature on the development of CBCT interventions be included?<br>[7] the details for 'setting' need justification, relating back to my point about defining CBCT - are you excluding programs in medical settings because these are not community-based? in which case, make it clear what your definition is.<br>[8] It looks like studies focused on prevention will not be included? If so, make this clearer.<br>[9] it is good practice to also search grey literature, please consider this.<br>[10] consider using TIDIER guidelines to describe the interventions included in your review. |

**VERSION 1 – AUTHOR RESPONSE**

Review 1's Comments to Author:

The systematic review protocol title "Community-Based Culturally Tailored Education Programs for Black Adults with Cardiovascular Disease, Diabetes, Hypertension, and Stroke: A Systematic Review Protocol" is well formulated and it is informative. The aim of the proposed systematic review is clear and relevant. The background has been provided by adequate literature review including recent relevant studies and systematic reviews, as well. The authors clearly state the gap in the research which they are looking to fill rising clearly defined research questions. The study design is appropriate to answer two review questions stated in the protocol, and applied methods are described sufficiently to assure repeatability.

This systematic review will improve understanding on the design, structure and definition of community-based culturally tailored education programs (CBCT) that have been used between 2000-2021 to improve health outcomes in Black adults with cardiovascular disease, hypertension, diabetes,

and stoke that is important for assuring more efficiency of future CBCT programs using Kreuter and colleagues framework.
Author's response: Thank you for recognizing the importance of our study and providing us with this positive feedback.

Review 2's Comments to Author:
[1] The title is incomplete and should include mention of the types of evidence being evaluated and the outcomes of interest.
Author's response: As suggested, we have revised the title to include a mention of the types of evidence being evaluated and the outcomes of interest (Page 1): Community-Based Culturally Tailored Education Programs for Black Adults with Cardiovascular Disease, Diabetes, Hypertension, and Stroke: A Systematic Review Protocol of Primary Empirical Studies

[2] Abstract - the authors make some strong claims in the abstract about the role of systemic racism and disparities in care. Whilst these are undeniably important drivers of health inequalities, there are a broad range of additional factors that contribute so i would encourage the authors to temper these statements a little, acknowledging that these are in part responsible.
Author's response: Thank you for your feedback. As suggested, we have revised the introduction to acknowledge that these factors are in part responsible for health inequalities and not the only factors (page 2): "Chronic conditions and stroke disproportionately affect Black adults in communities all around the world partly due to patterns of systemic racism, disparities in care, and lack of resources. Culturally-tailored programs can potentially meet the needs of the communities they serve, including Black adults who may experience reduced access to postacute services." We have also reviewed the manuscript to ensure that this is reflected in the remaining sections of the protocol.

[3] Title and methods - the review is focused on 'community-based' programs but how are they defining 'community-based' and how does this working definition compare/differ to definitions used by others in this field. This is a phrase that is used broadly and means different things to different people. I have the same comment regarding 'culturally tailored' - are the authors considering that 'culturally tailored' is distinct from 'community-based' i.e. could they have programs that are culturally tailored but not community-based and vice versa? Clear working definitions needed here.
Author's response: We recognize the importance of defining the terms 'community-based' and 'culturally tailored,' considering the emphasis on both of these terms for the systematic review and the varying definitions used for each term. The definitions have been added to reflect your suggestions (Page 5-6). For this review, a 'community-based' refers to an intervention delivered within a community setting (e.g. an individual's home, healthcare clinic[23]) that brings people together with the intention of empowering members of the community to share their own knowledge and experience to develop a common understanding to address common problems faced by communities.[24] Multiple terms have been used, sometimes interchangeably, to describe interventions that take culture (e.g. "culturally appropriate," "culturally targeted," and "culturally relevant"); in this review culturally tailored is used to refer to 'recognition of a group's cultural values, beliefs and behaviors' as indispensable context to addressing specific health problems impacting a specific ethnic community.[25] The combination of community-based and culturally tailored allows for the empowerment of individuals living within communities to use their cultural values, beliefs, and behaviors to inform programs and interventions to address health issues in their community.
The inclusion criteria for this study state that the education programs must be community-based and culturally tailored. While there may be overlap with the inclusion of people from target communities, this doesn't mean the programs would be culturally tailored.

[4] There are several frameworks available to evaluate culturally tailored strategies, can the authors briefly justify why they have chosen Kreuter's over the other options.

Author's response: We have added additional information to validate our choice to use the Krueter et al. framework and acknowledged the existence of other frameworks that have overlap but were determined to be insufficient for this proposed work (Page 6) . Multiple frameworks [26–28] exist to evaluate and develop interventions based on the cultural characteristics of the communities the interventions are meant to serve. While many of these frameworks outline similar approaches, concepts and ideas, the culturally appropriate framework developed by Kreuter et al. provides evaluation tools for surface level cultural characteristics (e.g. materials), deep structure cultural characteristics (e.g.language) and socio-cultural issues facing target communities.[16]

[5] Methods - the authors state 'the studies will be limited to articles published in English language'. Can they be clear if the SEARCH was limited or whether non-English studies were included in the search but excluded in the screening? Limiting the search has a very big effect, it is best to not limit the search.
Author's response: As suggested, we have specified that there were no language restrictions when implementing the search strategy, however, we will exclude non-English studies at the screening phase of the review.
-We have indicated that language restrictions were not a part of the search strategy in the methods (Page 8): "No language restrictions will be applied to the search strategy."
-We have acknowledged this as a limitation of our review (page 14): "Fourth, for feasibility and resource constraints, we decided to limit inclusion of articles to those available in English (i.e. articles will be excluded during title/abstract screening); however we recognize that we may miss potentially relevant literature."

[6] Top of page 10 is the following statement: 'we will include published empirical studies of any study design that gather evidence from a CBCT program..' It is not clear what sort of literature will be excluded - do the studies need to be an evaluation of a program on clinical outcomes or will literature on the development of CBCT interventions be included?
Author's response: Thank you for providing us with an opportunity to clarify this. We have further clarified details of the type of literature that will be included and excluded:
-We have added our working definition of empirical studies in the protocol (Page 9): "Empirical studies are any study "obtaining direct, observable information from the world, rather than, for example, by theorising, or by reasoning, or by arguing from first principles."[34]"
-the programs to be included in this review must measure the outcomes across the intervention. Published literature that only details the development of interventions will be excluded from the review (Page 9-10). "The key concept from this definition is the "observable information from the world" that is found from empirical studies (e.g. pilot studies, randomized control trials, cohort studies, qualitative interviews, surveys) and not from opinion letters, commentaries protocols, or intervention descriptions."[35]"

[7] the details for 'setting' need justification, relating back to my point about defining CBCT - are you excluding programs in medical settings because these are not community-based? in which case, make it clear what your definition is.
Author's response: Thank you for pointing out the need to further clarify what we mean by setting. A need exists for more effective models and frameworks for community-based interventions, including definitions of what constitutes 'community-based' (Page 5). "For this review, a 'community-based' refers to an intervention delivered within a community setting (e.g. an individual's home, healthcare clinic)[23] that brings people together with the intention of empowering members of the community to share their own knowledge and experience to develop a common understanding to address common problems faced by communities.[24]"

[8] It looks like studies focused on prevention will not be included? If so, make this clearer.

Author's response: This is accurate. To further enhance clarity for readers, we have explicitly specified that studies focused on prevention will not be included in this systematic review given the review purpose is to determine what CBCT programs have been developed to address individuals who are already experiencing the target conditions: (page 10): "This review will include programs which deliver culturally-tailored education to the Black adults with cardiovascular disease, hypertension, diabetes, or stroke. Thus, studies focused on prevention will not be included in this systematic review."
-Limitations (Page 14): "Fifth, the focus of this review will be on interventions designed and developed for people living with the target condition rather than prevention. Future research should explore studies that focus specifically on culturally tailored prevention programs and interventions."

[9] it is good practice to also search grey literature, please consider this.
Author's response: Thank you for encouraging us to consider gray literature. We have further explained our decision for excluding gray literature within our review (page 10): "Our research team decided to limit to empirical published studies to effectively address our research aims. We have excluded gray literature due to lack of peer-review and the variable quality and inconsistent reporting of program outcomes."
-Limitations (page 14): While gray literature is not deemed necessary to answer our questions, omitting gray literature may introduce publication bias,[43] and it could be a valuable resource for future research (e.g. informing intervention development).

[10] consider using TIDIER guidelines to describe the interventions included in your review.
Author's response: Thank you for this suggestion and we agree the TIDIER guidelines would be valuable to consider. As such, we have decided to customize our data extraction form based on the Joanna Briggs Institute Manual for Evidence Synthesis, Kreuter's five strategies and TIDIER (page 12): "A customized data extraction form will be created based on the Joanna Briggs Institute (JBI) Manual for Evidence Synthesis,[37] TIDIER guidelines to identify essential program characteristics (Tidier guidelines),[39] and Kreuter's five strategies of culturally appropriate interventions to determine the number and type of strategies used for each program included in the review.[16]"